

Homework 1

SI252 Reinforcement Learning

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Problem 1. Random variables

A random variable turns the result of an experiment into a number, so we can calculate probabilities and averages. For example, let X be the number of heads in two coin tosses:

$$X(HH) = 2, \quad X(HT) = X(TH) = 1, \quad X(TT) = 0.$$

X is a function because it assigns a number to each outcome. The rule is fixed, but we do not know which outcome will occur, so the value of X is uncertain. This is why it is called a *random variable*.

Problem 2. Probability and estimation identities

(a)

1. Both X and Y are discrete

By the definition of conditional probability,

$$P(Y = y \mid X = x) = \frac{P(X = x, Y = y)}{P(X = x)}.$$

Also,

$$P(X = x, Y = y) = P(X = x \mid Y = y)P(Y = y).$$

Substituting gives

$$P(Y = y \mid X = x) = \frac{P(X = x \mid Y = y)P(Y = y)}{P(X = x)}.$$

2. X is discrete and Y is continuous

Fix x such that $P(X = x) > 0$. For every measurable set $B \subseteq \mathbb{R}$, conditioning on Y yields

$$P(X = x, Y \in B) = \int_B P(X = x \mid Y = y) f_Y(y) dy.$$

Hence,

$$\begin{aligned} P(Y \in B \mid X = x) &= \frac{P(X = x, Y \in B)}{P(X = x)} \\ &= \int_B \frac{P(X = x \mid Y = y)f_Y(y)}{P(X = x)} dy. \end{aligned}$$

By the definition of a conditional density, the integrand is a density of Y conditional on $X = x$. Therefore,

$$f_{Y|X}(y \mid x) = \frac{P(X = x \mid Y = y)f_Y(y)}{P(X = x)}.$$

3. X is continuous and Y is discrete

Fix y such that $P(Y = y) > 0$. For every measurable set $A \subseteq \mathbb{R}$,

$$\begin{aligned} P(X \in A, Y = y) &= P(Y = y)P(X \in A \mid Y = y) \\ &= \int_A f_{X|Y}(x \mid y)P(Y = y) dx. \end{aligned}$$

On the other hand, conditioning on X gives

$$P(X \in A, Y = y) = \int_A P(Y = y \mid X = x)f_X(x) dx.$$

Since these integrals are equal for every measurable set A , their integrands are equal almost everywhere:

$$P(Y = y \mid X = x)f_X(x) = f_{X|Y}(x \mid y)P(Y = y).$$

Dividing by $f_X(x) > 0$, we obtain

$$P(Y = y \mid X = x) = \frac{f_{X|Y}(x \mid y)P(Y = y)}{f_X(x)}.$$

4. Both X and Y are continuous and admit a joint density

Let $f_{X,Y}(x, y)$ denote the joint density. By the definition of conditional density,

$$f_{Y|X}(y \mid x) = \frac{f_{X,Y}(x, y)}{f_X(x)}.$$

Moreover,

$$f_{X,Y}(x, y) = f_{X|Y}(x \mid y)f_Y(y).$$

Substituting yields

$$f_{Y|X}(y \mid x) = \frac{f_{X|Y}(x \mid y)f_Y(y)}{f_X(x)}.$$

(b)

1. Both X and Y are discrete

The events $\{Y = y\}$, taken over the possible values of Y , form a countable partition of the sample space up to a null set. Therefore, by countable additivity,

$$P(X = x) = \sum_y P(X = x, Y = y).$$

Using the definition of conditional probability,

$$P(X = x, Y = y) = P(X = x | Y = y)P(Y = y).$$

Consequently,

$$P(X = x) = \sum_y P(X = x | Y = y)P(Y = y).$$

The sum may be restricted to values y for which $P(Y = y) > 0$.

2. X is discrete and Y is continuous

Fix x such that $P(X = x) > 0$. By the previously established Bayes' rule,

$$f_{Y|X}(y | x) = \frac{P(X = x | Y = y)f_Y(y)}{P(X = x)}.$$

Rearranging gives

$$P(X = x | Y = y)f_Y(y) = P(X = x)f_{Y|X}(y | x).$$

Integrating both sides with respect to y , we obtain

$$\begin{aligned} \int_{-\infty}^{\infty} P(X = x | Y = y)f_Y(y) dy &= P(X = x) \int_{-\infty}^{\infty} f_{Y|X}(y | x) dy \\ &= P(X = x). \end{aligned}$$

3. X is continuous and Y is discrete

Let $A \subseteq \mathbb{R}$ be any measurable set. Applying the discrete law of total probability to the event $\{X \in A\}$, we have

$$P(X \in A) = \sum_y P(X \in A | Y = y)P(Y = y).$$

By the definition of conditional density,

$$P(X \in A | Y = y) = \int_A f_{X|Y}(x | y) dx.$$

Thus,

$$\begin{aligned} P(X \in A) &= \sum_y \left[\int_A f_{X|Y}(x | y) dx \right] P(Y = y) \\ &= \int_A \left[\sum_y f_{X|Y}(x | y)P(Y = y) \right] dx. \end{aligned}$$

The interchange of the sum and the integral is justified by Tonelli's theorem, since all the terms are nonnegative.

On the other hand,

$$P(X \in A) = \int_A f_X(x) dx.$$

Since these expressions agree for every measurable set A , uniqueness of densities yields

$$f_X(x) = \sum_y f_{X|Y}(x | y)P(Y = y).$$

4. Both X and Y are continuous and admit a joint density

Assume that X and Y have a joint density $f_{X,Y}(x, y)$. By marginalization,

$$f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x, y) dy.$$

The conditional density satisfies

$$f_{X,Y}(x, y) = f_{X|Y}(x | y)f_Y(y)$$

almost everywhere. Substituting gives

$$f_X(x) = \int_{-\infty}^{\infty} f_{X|Y}(x | y)f_Y(y) dy.$$

(c)

$$\text{Var}(Y | X) = E[Y^2 | X] - (E[Y | X])^2.$$

Taking expectations and applying the law of total expectation gives

$$\begin{aligned} E[\text{Var}(Y | X)] &= E[E[Y^2 | X]] - E[(E[Y | X])^2] \\ &= E[Y^2] - E[(E[Y | X])^2]. \end{aligned}$$

Using the variance identity $\text{Var}(Z) = E[Z^2] - (E[Z])^2$, we obtain

$$\begin{aligned} \text{Var}(E[Y | X]) &= E[(E[Y | X])^2] - (E[E[Y | X]])^2 \\ &= E[(E[Y | X])^2] - (E[Y])^2, \end{aligned}$$

where the last equality follows from the law of total expectation.

Adding the expressions above, the middle terms cancel:

$$\begin{aligned} &E[\text{Var}(Y | X)] + \text{Var}(E[Y | X]) \\ &= E[Y^2] - E[(E[Y | X])^2] + E[(E[Y | X])^2] - (E[Y])^2 \\ &= E[Y^2] - (E[Y])^2 \\ &= \text{Var}(Y). \end{aligned}$$

Therefore,

$$\text{Var}(Y) = E[\text{Var}(Y | X)] + \text{Var}(E[Y | X]).$$

(d)

Assume that $E[X^2] < \infty$, $E[Y^2] < \infty$, and $\text{Var}(X) > 0$. We seek constants a and b that minimize

$$Q(a, b) = E[(Y - a - bX)^2].$$

$$Q(a, b) = E[Y^2] - 2aE[Y] - 2bE[XY] + a^2 + 2abE[X] + b^2E[X^2].$$

$$\frac{\partial Q}{\partial a} = -2E[Y] + 2a + 2bE[X] = 0.$$

Hence,

$$a = E[Y] - bE[X].$$

Similarly,

$$\frac{\partial Q}{\partial b} = -2E[XY] + 2aE[X] + 2bE[X^2] = 0,$$

so

$$E[XY] = aE[X] + bE[X^2].$$

Substituting $a = E[Y] - bE[X]$, we obtain

$$E[XY] = E[X]E[Y] - b(E[X])^2 + bE[X^2].$$

Rearranging,

$$E[XY] - E[X]E[Y] = b(E[X^2] - (E[X])^2).$$

By the definitions of covariance and variance,

$$\text{Cov}(X, Y) = b \text{Var}(X).$$

Therefore,

$$b = \frac{\text{Cov}(X, Y)}{\text{Var}(X)}.$$

Thus,

$$L[Y | X] = E[Y] + \frac{\text{Cov}(X, Y)}{\text{Var}(X)}(X - E[X]).$$

The objective is a convex quadratic in (a, b) , with a positive-definite Hessian when $\text{Var}(X) > 0$, so this stationary point is the unique minimum. If $\text{Var}(X) = 0$, X is constant almost surely and the best predicted value is simply $E[Y]$.

(e)

Let $m(X) = \mathbb{E}[Y | X]$, and assume that Y has a finite second moment. For every measurable ϕ with $\mathbb{E}[\phi(X)^2] < \infty$, the tower property gives

$$\begin{aligned} \mathbb{E}[(Y - m(X))\phi(X)] &= \mathbb{E}[\mathbb{E}[(Y - m(X))\phi(X) | X]] \\ &= \mathbb{E}[\phi(X)(\mathbb{E}[Y | X] - m(X))] \\ &= 0. \end{aligned}$$

This proves the orthogonality property.

Conversely, assume that $g(X)$ has a finite second moment and that, for every such measurable function ϕ ,

$$\mathbb{E}[(Y - g(X))\phi(X)] = 0.$$

By the orthogonality property just proved,

$$\mathbb{E}[(Y - m(X))\phi(X)] = 0.$$

Subtracting the second equality from the first gives

$$\mathbb{E}[(m(X) - g(X))\phi(X)] = 0.$$

Since this holds for every such ϕ , we may choose

$$\phi(X) = m(X) - g(X),$$

which is square-integrable. Therefore,

$$\mathbb{E}[(m(X) - g(X))^2] = 0.$$

Since a nonnegative random variable with zero expectation is zero almost surely, $g(X) = m(X) = \mathbb{E}[Y | X]$ almost surely.

(f)

Proof. Let $\mu = \mathbb{E}[Z]$. By Chebyshev's inequality,

$$\mathbb{P}(|Z - \mu| \geq a) \leq \frac{\sigma^2}{a^2}.$$

We consider two cases:

- If $a^2 \leq \sigma^2$, then

$$\mathbb{P}(|Z - \mu| \geq a) \leq 1 \leq \frac{2\sigma^2}{\sigma^2 + a^2}.$$

- If $a^2 > \sigma^2$, then $\sigma^2 + a^2 \leq 2a^2$, and hence

$$\mathbb{P}(|Z - \mu| \geq a) \leq \frac{\sigma^2}{a^2} \leq \frac{2\sigma^2}{\sigma^2 + a^2}.$$

Therefore, for every $a > 0$,

$$\mathbb{P}(|Z - \mathbb{E}[Z]| \geq a) \leq \frac{2\sigma^2}{\sigma^2 + a^2}.$$

(g)

Proof. Let

$$Y_k = X_k - \mathbb{E}[X_k], \quad V = \sum_{k=1}^n (b_k - a_k)^2.$$

Then the Y_k are independent, $\mathbb{E}[Y_k] = 0$, and

$$a_k - \mathbb{E}[X_k] \leq Y_k \leq b_k - \mathbb{E}[X_k].$$

Thus, by Hoeffding's lemma, for every $\lambda > 0$,

$$\mathbb{E}[e^{\lambda Y_k}] \leq \exp\left(\frac{\lambda^2 (b_k - a_k)^2}{8}\right).$$

Using independence, we obtain

$$\begin{aligned} \mathbb{E}[e^{\lambda(S_n - \mu)}] &= \prod_{k=1}^n \mathbb{E}[e^{\lambda Y_k}] \\ &\leq \exp\left(\frac{\lambda^2 V}{8}\right). \end{aligned}$$

Assume $V > 0$ and $t > 0$. By Markov's inequality, for every $\lambda > 0$,

$$\begin{aligned} \mathbb{P}(S_n - \mu \geq t) &= \mathbb{P}(e^{\lambda(S_n - \mu)} \geq e^{\lambda t}) \\ &\leq e^{-\lambda t} \mathbb{E}[e^{\lambda(S_n - \mu)}] \\ &\leq \exp\left(-\lambda t + \frac{\lambda^2 V}{8}\right). \end{aligned}$$

Choosing $\lambda = 4t/V$ gives

$$\mathbb{P}(S_n - \mu \geq t) \leq \exp\left(-\frac{2t^2}{V}\right).$$

Applying the same argument to the independent random variables $-X_1, \dots, -X_n$, whose interval lengths remain $b_k - a_k$, yields

$$\mathbb{P}(S_n - \mu \leq -t) = \mathbb{P}(\mu - S_n \geq t) \leq \exp\left(-\frac{2t^2}{V}\right).$$

Therefore, by the union bound,

$$\begin{aligned} \mathbb{P}(|S_n - \mu| \geq t) &\leq \mathbb{P}(S_n - \mu \geq t) + \mathbb{P}(S_n - \mu \leq -t) \\ &\leq 2 \exp\left(-\frac{2t^2}{V}\right). \end{aligned}$$

For $t = 0$, the inequality holds trivially since $1 \leq 2$. Hence, when $V > 0$,

$$\mathbb{P}(|S_n - \mu| \geq t) \leq 2 \exp\left(-\frac{2t^2}{\sum_{k=1}^n (b_k - a_k)^2}\right), \quad t \geq 0.$$

If $V = 0$, all X_k are constant almost surely, so $S_n = \mu$ almost surely. In this degenerate case, the tail probability is 0 for $t > 0$ and 1 for $t = 0$.

Problem 3. Exponential conditional expectations

Assume that $X_i \sim \text{Exp}(\lambda_i)$, where λ_i is the rate parameter. Then

$$f_{X_i}(x) = \lambda_i e^{-\lambda_i x}, \quad x > 0.$$

By the memoryless property of the exponential distribution,

$$X_i - t \mid X_i > t \sim \text{Exp}(\lambda_i).$$

Therefore,

$$E(X_1 \mid X_1 > 2026) = 2026 + E(X_1) = \boxed{2026 + \frac{1}{\lambda_1}}.$$

For the second term,

$$E(X_1 \mid X_1 < 2026) = \frac{\int_0^{2026} x \lambda_1 e^{-\lambda_1 x} dx}{P(X_1 < 2026)}.$$

Evaluating the integral gives

$$E(X_1 \mid X_1 < 2026) = \frac{1}{\lambda_1} - \frac{2026 e^{-2026 \lambda_1}}{1 - e^{-2026 \lambda_1}}$$

or, equivalently,

$$E(X_1 \mid X_1 < 2026) = \frac{1}{\lambda_1} - \frac{2026}{e^{2026 \lambda_1} - 1}$$

Finally, since X_1, X_2, X_3 are independent, we can apply the memoryless property to each variable:

$$\begin{aligned} E(X_1 + X_2 + X_3 \mid X_1 > 1997, X_2 > 2014, X_3 > 2026) \\ = \left(1997 + \frac{1}{\lambda_1}\right) + \left(2014 + \frac{1}{\lambda_2}\right) + \left(2026 + \frac{1}{\lambda_3}\right). \end{aligned}$$

Hence,

$$E(X_1 + X_2 + X_3 \mid X_1 > 1997, X_2 > 2014, X_3 > 2026) = 6037 + \frac{1}{\lambda_1} + \frac{1}{\lambda_2} + \frac{1}{\lambda_3}$$

Problem 4. Bivariate normal construction

Let Z_1, Z_2 be independent standard normal random variables, and define

$$X = \sigma_X Z_1 + \mu_X, \quad Y = \sigma_Y \left(\rho Z_1 + \sqrt{1 - \rho^2} Z_2 \right) + \mu_Y,$$

where $\sigma_X, \sigma_Y > 0$ and $-1 < \rho < 1$.

(a) For any $a, b \in \mathbb{R}$,

$$aX + bY = a\mu_X + b\mu_Y + (a\sigma_X + b\rho\sigma_Y)Z_1 + b\sigma_Y\sqrt{1 - \rho^2}Z_2.$$

Since a linear combination of independent normal random variables is normal, $aX + bY$ is normal. Therefore, X and Y are jointly normal.

(b) We have

$$\text{Var}(X) = \sigma_X^2, \quad \text{Var}(Y) = \sigma_Y^2(\rho^2 + 1 - \rho^2) = \sigma_Y^2.$$

Using independence,

$$\begin{aligned} \text{Cov}(X, Y) &= \sigma_X \sigma_Y \left[\rho \text{Var}(Z_1) + \sqrt{1 - \rho^2} \text{Cov}(Z_1, Z_2) \right] \\ &= \rho \sigma_X \sigma_Y. \end{aligned}$$

Hence,

$$\text{Corr}(X, Y) = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y} = \rho$$

(c) The inverse transformation is

$$z_1 = \frac{x - \mu_X}{\sigma_X}, \quad z_2 = \frac{\frac{y - \mu_Y}{\sigma_Y} - \rho \frac{x - \mu_X}{\sigma_X}}{\sqrt{1 - \rho^2}}.$$

Its Jacobian determinant has absolute value

$$\left| \frac{\partial(z_1, z_2)}{\partial(x, y)} \right| = \frac{1}{\sigma_X \sigma_Y \sqrt{1 - \rho^2}}.$$

Since

$$f_{Z_1, Z_2}(z_1, z_2) = \frac{1}{2\pi} \exp\left(-\frac{z_1^2 + z_2^2}{2}\right),$$

the change-of-variables formula gives

$$\begin{aligned} f_{X, Y}(x, y) &= \frac{1}{2\pi \sigma_X \sigma_Y \sqrt{1 - \rho^2}} \\ &\times \exp\left\{-\frac{1}{2(1 - \rho^2)} \left[\frac{(x - \mu_X)^2}{\sigma_X^2} - \frac{2\rho(x - \mu_X)(y - \mu_Y)}{\sigma_X \sigma_Y} + \frac{(y - \mu_Y)^2}{\sigma_Y^2} \right]\right\}, \end{aligned}$$

for all $x, y \in \mathbb{R}$.

Problem 5. A three-state Markov chain

The transition matrix, with states ordered as 1, 2, 3, is

$$P = \begin{pmatrix} \frac{1}{4} & 0 & \frac{3}{4} \\ \frac{1}{2} & 0 & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{4} & \frac{1}{4} \end{pmatrix},$$

where

$$p_{ij} = P(X_{n+1} = j \mid X_n = i).$$

(a)

Since the chain is time-homogeneous, the one-step transition probabilities do not depend on time. Therefore,

$$P(X_5 = 3 \mid X_4 = 2) = p_{23} = \frac{1}{2},$$

and

$$P(X_4 = 1 \mid X_3 = 2) = p_{21} = \frac{1}{2}.$$

(b)

Let $\boldsymbol{\pi} = (\pi_1, \pi_2, \pi_3)$ be the stationary distribution. It satisfies

$$\boldsymbol{\pi}P = \boldsymbol{\pi}, \quad \pi_1 + \pi_2 + \pi_3 = 1.$$

Thus,

$$\begin{cases} \pi_1 = \frac{1}{4}\pi_1 + \frac{1}{2}\pi_2 + \frac{1}{2}\pi_3, \\ \pi_2 = \frac{1}{4}\pi_3, \\ \pi_3 = \frac{3}{4}\pi_1 + \frac{1}{2}\pi_2 + \frac{1}{4}\pi_3. \end{cases}$$

Substituting $\pi_2 = \frac{1}{4}\pi_3$ into the first equation gives

$$\frac{3}{4}\pi_1 = \frac{1}{8}\pi_3 + \frac{1}{2}\pi_3 = \frac{5}{8}\pi_3,$$

so

$$\pi_1 = \frac{5}{6}\pi_3.$$

Using the normalization condition,

$$\left(\frac{5}{6} + \frac{1}{4} + 1\right)\pi_3 = 1,$$

which yields

$$\pi_3 = \frac{12}{25}, \quad \pi_2 = \frac{3}{25}, \quad \pi_1 = \frac{2}{5}.$$

Hence,

$$\boldsymbol{\pi} = \left(\frac{2}{5}, \frac{3}{25}, \frac{12}{25}\right).$$

(c)

The period of state i is defined by

$$d(i) = \gcd\{n \geq 1 : (P^n)_{ii} > 0\}.$$

Starting from state 2, the following return paths have positive probability:

$$2 \rightarrow 3 \rightarrow 2 \quad \text{and} \quad 2 \rightarrow 3 \rightarrow 3 \rightarrow 2.$$

Thus, returns are possible in both two and three steps. Since $\gcd(2, 3) = 1$,

$$d(2) = 1.$$

All three states communicate, so the chain is irreducible and all states have the same period. Therefore, the period of the whole chain is 1.

(d)

By the multiplication rule and the Markov property,

$$\begin{aligned} P(X_0 = 3, X_1 = 2, X_2 = 1) &= P(X_0 = 3) P(X_1 = 2 \mid X_0 = 3) P(X_2 = 1 \mid X_1 = 2) \\ &= \frac{1}{5} \cdot \frac{1}{4} \cdot \frac{1}{2} \\ &= \frac{1}{40}. \end{aligned}$$

(e)

Since there are two transitions from time 3 to time 5,

$$P(X_5 = j \mid X_3 = 1) = (P^2)_{1j} = \sum_{k=1}^3 p_{1k} p_{kj}.$$

Consequently,

$$P(X_5 = 1 \mid X_3 = 1) = \frac{1}{4} \cdot \frac{1}{4} + \frac{3}{4} \cdot \frac{1}{2} = \frac{7}{16},$$

$$P(X_5 = 2 \mid X_3 = 1) = \frac{1}{4} \cdot 0 + \frac{3}{4} \cdot \frac{1}{4} = \frac{3}{16},$$

$$P(X_5 = 3 \mid X_3 = 1) = \frac{1}{4} \cdot \frac{3}{4} + \frac{3}{4} \cdot \frac{1}{4} = \frac{6}{16}.$$

The conditional expectation is

$$\begin{aligned} E(X_5 \mid X_3 = 1) &= 1 \cdot \frac{7}{16} + 2 \cdot \frac{3}{16} + 3 \cdot \frac{6}{16} \\ &= \frac{31}{16}. \end{aligned}$$

The conditional second moment is

$$\begin{aligned} E(X_5^2 \mid X_3 = 1) &= 1^2 \cdot \frac{7}{16} + 2^2 \cdot \frac{3}{16} + 3^2 \cdot \frac{6}{16} \\ &= \frac{73}{16}. \end{aligned}$$

Therefore,

$$\begin{aligned} \text{Var}(X_5 \mid X_3 = 1) &= E(X_5^2 \mid X_3 = 1) - [E(X_5 \mid X_3 = 1)]^2 \\ &= \frac{73}{16} - \left(\frac{31}{16}\right)^2 \\ &= \frac{207}{256}. \end{aligned}$$

Problem 6. Estimating a coin's head probability

Assume $n \geq 1$. Given p , the number of heads satisfies

$$X \mid p \sim \text{Binomial}(n, p).$$

Hence, for the observation $X = k$,

$$P(X = k \mid p) = \binom{n}{k} p^k (1-p)^{n-k}, \quad 0 \leq k \leq n.$$

(a) Maximum Likelihood Estimation (MLE)

Treating p as an unknown constant, the likelihood function is

$$L(p) = \binom{n}{k} p^k (1-p)^{n-k}, \quad 0 \leq p \leq 1.$$

For $0 < k < n$, the log-likelihood is

$$\ell(p) = \log \binom{n}{k} + k \log p + (n-k) \log(1-p).$$

Differentiating and setting the derivative equal to zero gives

$$\ell'(p) = \frac{k}{p} - \frac{n-k}{1-p} = 0,$$

so

$$k(1-p) = (n-k)p \implies p = \frac{k}{n}.$$

Moreover,

$$\ell''(p) = -\frac{k}{p^2} - \frac{n-k}{(1-p)^2} < 0,$$

so this point maximizes the likelihood.

If $k = 0$, the likelihood is maximized at $p = 0$; if $k = n$, it is maximized at $p = 1$. Thus, in all cases,

$$\hat{p}_{\text{MLE}} = \frac{k}{n}.$$

(b) Maximum A Posteriori Estimation (MAP)

The prior distribution is

$$p \sim \text{Beta}(1, 1),$$

which is the uniform distribution on $[0, 1]$. Therefore, the prior density is $f(p) = 1$ on this interval.

By Bayes' rule,

$$f(p \mid X = k) \propto P(X = k \mid p) f(p) \propto p^k (1-p)^{n-k}.$$

Consequently,

$$p \mid X = k \sim \text{Beta}(k+1, n-k+1).$$

The MAP estimate maximizes the posterior density. Since the prior density is constant, maximizing the posterior density is equivalent to maximizing the likelihood. Hence,

$$\hat{p}_{\text{MAP}} = \frac{k}{n},$$

including the boundary cases $k = 0$ and $k = n$.

(c) Minimum Mean Squared Error Estimation (MMSE)

The MMSE estimate is the posterior mean:

$$\hat{p}_{\text{MMSE}} = E[p \mid X = k].$$

Since

$$\text{Unif}(0, 1) = \text{Beta}(1, 1),$$

the posterior distribution is again

$$p \mid X = k \sim \text{Beta}(k + 1, n - k + 1).$$

The mean of a $\text{Beta}(\alpha, \beta)$ distribution is $\alpha/(\alpha + \beta)$. Therefore,

$$\hat{p}_{\text{MMSE}} = \frac{k + 1}{n + 2}.$$

(d) Linear Least Squares Estimation (LLSE)

Using the given formula with $Y = p$, we obtain

$$\hat{p}_{\text{LLSE}} = E[p] + \frac{\text{Cov}(X, p)}{\text{Var}(X)}(k - E[X]).$$

All moments below are computed under the joint distribution specified by the prior and the observation model.

Since $p \sim \text{Unif}(0, 1)$,

$$E[p] = \frac{1}{2}, \quad E[p^2] = \frac{1}{3}, \quad \text{Var}(p) = \frac{1}{12}.$$

Also, $E[X \mid p] = np$, so the law of total expectation gives

$$E[X] = E[E[X \mid p]] = nE[p] = \frac{n}{2}.$$

By conditioning on p ,

$$E[Xp] = E[pE[X \mid p]] = nE[p^2] = \frac{n}{3}.$$

Thus,

$$\text{Cov}(X, p) = E[Xp] - E[X]E[p] = \frac{n}{3} - \frac{n}{2} \cdot \frac{1}{2} = \frac{n}{12}.$$

By the law of total variance,

$$\begin{aligned} \text{Var}(X) &= E[\text{Var}(X \mid p)] + \text{Var}(E[X \mid p]) \\ &= E[np(1 - p)] + \text{Var}(np) \\ &= n \left(\frac{1}{2} - \frac{1}{3} \right) + n^2 \left(\frac{1}{12} \right) \\ &= \frac{n(n + 2)}{12}. \end{aligned}$$

Therefore,

$$\begin{aligned}\hat{p}_{\text{LLSE}} &= \frac{1}{2} + \frac{n/12}{n(n+2)/12} \left(k - \frac{n}{2}\right) \\ &= \frac{1}{2} + \frac{k - n/2}{n+2} \\ &= \frac{k+1}{n+2}.\end{aligned}$$

Problem 7. Gaussian MMSE and LLSE

Method 1

Let

$$\mu_X = E[X], \quad \mu_Y = E[Y], \quad a = \frac{\text{Cov}(X, Y)}{\text{Var}(X)},$$

where we assume that $\text{Var}(X) > 0$. Define

$$Z = Y - \mu_Y - a(X - \mu_X).$$

Then

$$Y = \mu_Y + a(X - \mu_X) + Z.$$

First, the residual Z has mean zero:

$$\begin{aligned}E[Z] &= E[Y] - \mu_Y - a(E[X] - \mu_X) \\ &= 0.\end{aligned}$$

Moreover,

$$\begin{aligned}\text{Cov}(X, Z) &= E[(X - E[X])(Z - E[Z])] \\ &= E[(X - \mu_X)Z] \\ &= E[(X - \mu_X)((Y - \mu_Y) - a(X - \mu_X))] \\ &= E[(X - \mu_X)(Y - \mu_Y)] - aE[(X - \mu_X)^2] \\ &= \text{Cov}(X, Y) - a \text{Var}(X) \\ &= 0.\end{aligned}$$

Since X and Y are jointly Gaussian and Z is an affine function of X and Y , the variables X and Z are also jointly Gaussian. Thus, their zero covariance implies that they are independent. Consequently,

$$E[Z | X] = E[Z] = 0.$$

Taking conditional expectations gives

$$\begin{aligned}E[Y | X] &= E[\mu_Y + a(X - \mu_X) + Z | X] \\ &= \mu_Y + a(X - \mu_X) + E[Z | X] \\ &= \mu_Y + a(X - \mu_X).\end{aligned}$$

To verify the minimum mean squared error property, write

$$m(X) = \mu_Y + a(X - \mu_X).$$

Since $Y = m(X) + Z$, the mean squared error of $m(X)$ is

$$E[(Y - m(X))^2] = E[Z^2].$$

Any other square-integrable predictor can be written as

$$g(X) = m(X) + d(X).$$

Using $E[Z | X] = 0$, we obtain

$$E[Zd(X)] = E[d(X)E[Z | X]] = 0.$$

Therefore,

$$\begin{aligned} E[(Y - g(X))^2] &= E[(Z - d(X))^2] \\ &= E[Z^2] - 2E[Zd(X)] + E[d(X)^2] \\ &= E[Z^2] + E[d(X)^2] \\ &\geq E[Z^2]. \end{aligned}$$

Hence $m(X) = E[Y | X]$ is the MMSE predictor. Since $m(X)$ is also affine in X , it is the best linear least-squares predictor $L[Y | X]$. Thus,

$$E[Y | X] = L[Y | X] = E[Y] + \frac{\text{Cov}(X, Y)}{\text{Var}(X)}(X - E[X])$$

Method 2

Let

$$\mu_X = E[X], \quad \mu_Y = E[Y], \quad \sigma_X^2 = \text{Var}(X) > 0, \quad \sigma_Y^2 = \text{Var}(Y).$$

First assume $\sigma_Y > 0$. Define

$$U = \frac{X - \mu_X}{\sigma_X}, \quad V = \frac{Y - \mu_Y}{\sigma_Y}, \quad \rho = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}.$$

For $|\rho| < 1$, the joint density of U, V is

$$f_{U,V}(u, v) = \frac{1}{2\pi\sqrt{1-\rho^2}} \exp\left(-\frac{u^2 - 2\rho uv + v^2}{2(1-\rho^2)}\right).$$

Given $U = u$, treat u as a constant. Completing the square in v , we get

$$u^2 - 2\rho uv + v^2 = (v - \rho u)^2 + (1 - \rho^2)u^2.$$

Therefore,

$$f_{U,V}(u, v) = \frac{e^{-u^2/2}}{2\pi\sqrt{1-\rho^2}} \exp\left(-\frac{(v - \rho u)^2}{2(1-\rho^2)}\right).$$

Since $f_{V|U}(v | u) = f_{U,V}(u, v)/f_U(u)$, all factors independent of v can be absorbed into a constant. Thus,

$$f_{V|U}(v | u) \propto \exp\left(-\frac{(v - \rho u)^2}{2(1 - \rho^2)}\right).$$

This is a Gaussian density symmetric about $v = \rho u$. Hence its mean is

$$E[V | U = u] = \rho u.$$

To return to the original variables, note that

$$Y = \mu_Y + \sigma_Y V.$$

Also, $X = x$ is equivalent to $U = (x - \mu_X)/\sigma_X$. Therefore,

$$\begin{aligned} E[Y | X = x] &= \mu_Y + \sigma_Y E[V | X = x] \\ &= \mu_Y + \sigma_Y E\left[V \mid U = \frac{x - \mu_X}{\sigma_X}\right] \\ &= \mu_Y + \sigma_Y \rho \frac{x - \mu_X}{\sigma_X} \\ &= \mu_Y + \frac{\text{Cov}(X, Y)}{\text{Var}(X)}(x - \mu_X). \end{aligned}$$

The conditional expectation minimizes mean squared error among all predictors based on X . Since the expression above is affine in X , it is also the best linear predictor $L[Y | X]$. Thus,

$$E[Y | X] = L[Y | X] = E[Y] + \frac{\text{Cov}(X, Y)}{\text{Var}(X)}(X - E[X]).$$

Degenerate cases. If $\sigma_Y = 0$, then $Y = \mu_Y$ almost surely and the result is immediate. If $\sigma_X, \sigma_Y > 0$ and $|\rho| = 1$, then

$$\text{Var}(V - \rho U) = 1 - \rho^2 = 0,$$

so $V = \rho U$ almost surely, giving the same affine conditional mean. If $\text{Var}(X) = 0$, the displayed covariance ratio is undefined, but both the MMSE and LLSE predicted values are $E[Y]$.

Problem 8. Expected return of the uniform-sum program

Let $X_1, X_2, \dots$ be independent random variables, each uniformly distributed on $(0, 1)$. Define

$$S_n = \sum_{i=1}^n X_i, \quad N = \min\{n \geq 1 : S_n > 1\}.$$

Here, N is the number of loop iterations, and the program returns S_N .

For a positive integer-valued random variable N , the definition of expectation gives

$$\mathbb{E}[N] = \Pr(N = 1) + 2\Pr(N = 2) + 3\Pr(N = 3) + \dots$$

Splitting each coefficient into a sum of ones and regrouping the nonnegative terms, we obtain

$$\begin{aligned}
 \mathbb{E}[N] &= [\Pr(N = 1) + \Pr(N = 2) + \Pr(N = 3) + \cdots] \\
 &\quad + [\Pr(N = 2) + \Pr(N = 3) + \cdots] \\
 &\quad + [\Pr(N = 3) + \cdots] + \cdots \\
 &= \sum_{i=1}^{\infty} \Pr(N \geq i) \\
 &= \sum_{n=0}^{\infty} \Pr(N > n).
 \end{aligned}$$

Since the increments are nonnegative,

$$\Pr(N > n) = \Pr(S_n \leq 1) = \frac{1}{n!},$$

where, for $n \geq 1$, the last equality follows from the volume of the region

$$\left\{ (x_1, \dots, x_n) : x_i \geq 0, \sum_{i=1}^n x_i \leq 1 \right\}.$$

For $n = 0$, the probability is $1 = 1/0!$. Therefore,

$$\mathbb{E}[N] = \sum_{n=0}^{\infty} \frac{1}{n!} = e.$$

We can derive the expected return value directly, without invoking Wald's equation. Define the actual contribution of the i th iteration by

$$Y_i = \begin{cases} X_i, & \text{if the } i\text{th iteration occurs,} \\ 0, & \text{otherwise.} \end{cases}$$

Equivalently,

$$Y_i = X_i \mathbf{1}_{\{N \geq i\}}.$$

The key observation is that whether the i th iteration occurs depends only on $X_1, \dots, X_{i-1}$, not on X_i . Hence the event $\{N \geq i\}$ is independent of X_i .

Since

$$\mathbb{E}[X_i] = \frac{1}{2},$$

we have

$$\mathbb{E}[Y_i] = \mathbb{E}[X_i] \Pr(N \geq i) = \frac{1}{2} \Pr(N \geq i).$$

The returned sum is

$$S_N = \sum_{i=1}^{\infty} Y_i.$$

All the terms are nonnegative, so we may interchange expectation and summation:

$$\begin{aligned}
 \mathbb{E}[S_N] &= \sum_{i=1}^{\infty} \mathbb{E}[Y_i] \\
 &= \frac{1}{2} \sum_{i=1}^{\infty} \Pr(N \geq i) \\
 &= \frac{1}{2} \mathbb{E}[N] \\
 &= \frac{e}{2} \approx 1.35914.
 \end{aligned}$$

Problem 9. Hoeffding's lemma

Suppose that $\mathbb{E}[X] = 0$ and $a \leq X \leq b$ almost surely. Then, for every $\lambda > 0$,

$$\mathbb{E}[e^{\lambda X}] \leq \exp\left(\frac{\lambda^2(b-a)^2}{8}\right).$$

If $a = b$, the zero-mean assumption implies that $X = 0$ almost surely, so the result is immediate. Henceforth, assume that $a < b$.

Since $\mathbb{E}[X] = 0$ and $a \leq X \leq b$, we have $a \leq 0 \leq b$. For every $x \in [a, b]$,

$$x = \frac{b-x}{b-a}a + \frac{x-a}{b-a}b.$$

The two coefficients are nonnegative and sum to one. Therefore, by the convexity of $x \mapsto e^{\lambda x}$,

$$e^{\lambda x} \leq \frac{b-x}{b-a}e^{\lambda a} + \frac{x-a}{b-a}e^{\lambda b}.$$

Substituting $x = X$, taking expectations, and using $\mathbb{E}[X] = 0$, we obtain

$$\mathbb{E}[e^{\lambda X}] \leq \frac{b}{b-a}e^{\lambda a} + \frac{-a}{b-a}e^{\lambda b}.$$

Let

$$p = \frac{-a}{b-a} \in [0, 1], \quad t = \lambda(b-a) > 0.$$

Then

$$1-p = \frac{b}{b-a}, \quad \lambda a = -pt, \quad \lambda b = (1-p)t.$$

Consequently,

$$\mathbb{E}[e^{\lambda X}] \leq (1-p)e^{-pt} + pe^{(1-p)t}.$$

Define

$$h(t) = \log\left((1-p)e^{-pt} + pe^{(1-p)t}\right) = -pt + \log(1-p + pe^t).$$

It suffices to prove that

$$h(t) \leq \frac{t^2}{8}.$$

First, observe that

$$h(0) = 0.$$

Differentiating once gives

$$h'(t) = -p + \frac{pe^t}{1 - p + pe^t},$$

and hence

$$h'(0) = 0.$$

Differentiating again gives

$$h''(t) = \frac{p(1-p)e^t}{(1-p+pe^t)^2}.$$

Let

$$q(t) = \frac{pe^t}{1-p+pe^t} \in [0, 1].$$

Then

$$h''(t) = q(t)(1 - q(t)).$$

For every $q \in [0, 1]$,

$$q(1 - q) = \frac{1}{4} - \left(q - \frac{1}{2}\right)^2 \leq \frac{1}{4}.$$

Therefore,

$$h''(t) \leq \frac{1}{4}.$$

Using $h'(0) = 0$, for every $s \geq 0$ we have

$$h'(s) = \int_0^s h''(u) du \leq \frac{s}{4}.$$

Using $h(0) = 0$ and integrating once more, we obtain

$$h(t) = \int_0^t h'(s) ds \leq \int_0^t \frac{s}{4} ds = \frac{t^2}{8}.$$

It follows that

$$\mathbb{E}[e^{\lambda X}] \leq e^{h(t)} \leq e^{t^2/8} = \exp\left(\frac{\lambda^2(b-a)^2}{8}\right).$$

Problem 10. Utility, fairness, and RL rewards

We use the constraint $x_1 + 2x_2 \leq 12$, with $x_1, x_2 \geq 0$. For logarithms, both rates must be positive.

(a)

A utility function gives a score to each outcome: a higher score means a better outcome. In symbols,

$$a \succeq b \iff u(a) \geq u(b).$$

Both $u(x) = x$ and $v(x) = \log x$ say that more service is better. But adding the two users' scores can give different answers. With equal weights,

Objective	Best allocation
$x_1 + x_2$	(12, 0)
$\log x_1 + \log x_2$	(6, 3)

The calculations are in part (c). The reason is that the two scores grow differently: their slopes are 1 and $1/x$. Under $\log x$, the same small increase in service helps a user with less service more. Knowing that each user wants more does not tell us whose extra service should count more. The utility functions and weights make that choice.

(b)

$u'(x) > 0$ means that more service gives more utility. $u''(x) < 0$ means that the slope gets smaller: each extra unit adds less utility than before. The curve rises but becomes flatter.

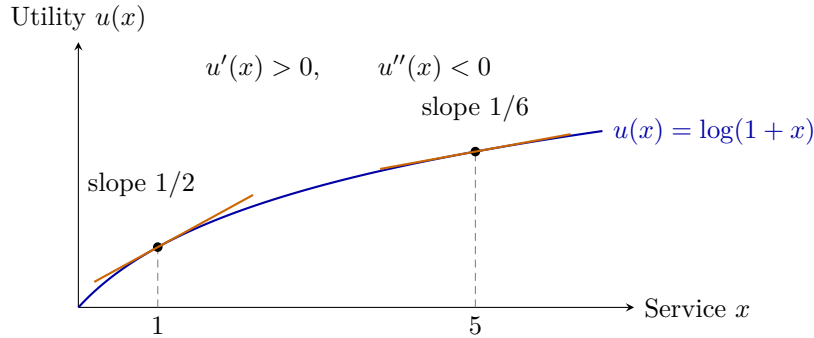


Figure 1: The curve rises but gets flatter: extra service adds less utility when a user already has more.

Why a transfer helps. Let $0 < a < b$ and $0 < \delta < (b - a)/2$. Define

$$h(t) = u(a + t) + u(b - t), \quad 0 \leq t \leq \delta.$$

Because $a + t < b - t$ and u' is decreasing,

$$h'(t) = u'(a + t) - u'(b - t) > 0.$$

Thus $h(\delta) > h(0)$, which is exactly

$$u(a + \delta) + u(b - \delta) > u(a) + u(b).$$

In words, the user with less service gains more utility than the other user loses.

Must the best rates be equal? No. The transfer above keeps total *service* fixed, but it may use more *resources*. For example, transferring service from user 1 to user 2 at $(6, 3)$ gives

$$(6 - \delta) + 2(3 + \delta) = 12 + \delta > 12,$$

which breaks the constraint. Even with the same strictly concave utility $\log x$, the best allocation is $(6, 3)$, not equal rates.

(c)

The three allocations.

1. **Total service.** Since

$$x_1 + x_2 \leq x_1 + 2x_2 \leq 12,$$

the maximum is 12, reached at $(12, 0)$.

2. **Logarithmic utility.** Maximizing the sum of the logarithms is the same as maximizing $x_1 x_2$. By AM–GM,

$$x_1 x_2 = \frac{x_1(2x_2)}{2} \leq \frac{1}{2} \left(\frac{x_1 + 2x_2}{2} \right)^2 \leq 18.$$

Equality needs $x_1 = 2x_2$ and $x_1 + 2x_2 = 12$, giving $(6, 3)$.

3. **Minimum service.** Let $m = \min\{x_1, x_2\}$. Then

$$3m \leq x_1 + 2x_2 \leq 12,$$

so $m \leq 4$, reached at $(4, 4)$.

The first choice gives the largest total service. The third gives equal service here. Log utility does neither; it gives *proportional fairness*, which compares percentage changes rather than equal amounts.

Proof of proportional fairness. Let $\mathbf{x}^*$ maximize $W(\mathbf{x}) = \sum_i \log x_i$ on the given convex set C . For any $\mathbf{y} \in C$, convexity means that

$$\mathbf{x}(t) = \mathbf{x}^* + t(\mathbf{y} - \mathbf{x}^*) \in C, \quad 0 \leq t \leq 1.$$

Set $g(t) = W(\mathbf{x}(t))$. Since $\mathbf{x}^*$ is best, $g(t) \leq g(0)$. Therefore its right derivative at zero cannot be positive:

$$0 \geq g'(0^+) = \sum_{i=1}^n \frac{y_i - x_i^*}{x_i^*}.$$

Each fraction is one user's relative change. For example, going from 4 to 5 gives $1/4$, or a 25% gain. The inequality says that the sum of percentage gains cannot exceed the sum of percentage losses. This is *proportional fairness*.

Weighting user 1 twice. Maximizing $2 \log x_1 + \log x_2$ is the same as maximizing $x_1^2 x_2$. Again, AM–GM gives

$$\begin{aligned} x_1^2 x_2 &= 2 \left(\frac{x_1}{2} \right) \left(\frac{x_1}{2} \right) (2x_2) \\ &\leq 2 \left(\frac{x_1 + 2x_2}{3} \right)^3 \leq 128. \end{aligned}$$

Equality needs $x_1/2 = 2x_2$ and $x_1 + 2x_2 = 12$. Hence the answer is $(8, 2)$. The weight 2 means that user 1's utility counts twice in the objective, not that user 1 must receive twice as much service.

(d)

The expected payoffs are equal:

$$\mathbb{E}[X_A] = 5, \quad \mathbb{E}[X_B] = \frac{1}{2}(1) + \frac{1}{2}(9) = 5.$$

So a person maximizing expected payoff values A and B equally. With log utility,

$$\mathbb{E}[\log X_A] = \log 5, \quad \mathbb{E}[\log X_B] = \frac{1}{2} \log 1 + \frac{1}{2} \log 9 = \log 3.$$

Now A is better. B's certainty equivalent is the sure amount c that has the same utility as B:

$$\log c = \log 3 \implies c = 3.$$

For any concave utility, Jensen's inequality gives

$$\mathbb{E}[u(X)] \leq u(\mathbb{E}[X]),$$

assuming these expectations are finite. Thus receiving the mean payoff for sure is at least as good as taking the gamble. This is *risk aversion*. Here we compare one person's possible payoffs; part (b) compares service given to different people. That is why one is about risk and the other is about fairness.

(e)

A lottery is a list of possible outcomes and their probabilities. For a finite set $\{z_1, \dots, z_n\}$, write a lottery as $p = (p_1, \dots, p_n)$, where $p_i \geq 0$ and $\sum_i p_i = 1$. Here $p \succeq q$ means that p is at least as good as q , and $p \sim q$ means they are equally good.

The VNM theorem uses four rules for comparing lotteries:

1. **Completeness:** any two lotteries can be compared: $p \succeq q$ or $q \succeq p$, or both.
2. **Transitivity:** if $p \succeq q$ and $q \succeq r$, then $p \succeq r$.
3. **Continuity:** if $p \succ q \succ r$, then for some $\alpha \in (0, 1)$,

$$q \sim \alpha p + (1 - \alpha)r.$$

In words, a suitable mix of the best and worst lotteries is as good as the middle one.

4. **Independence:** mixing both choices with the same third lottery in the same proportion keeps their order:

$$p \succeq q \iff \alpha p + (1 - \alpha)r \succeq \alpha q + (1 - \alpha)r,$$

for $0 < \alpha < 1$. The mixture chooses p with probability α and r otherwise, and similarly for q .

If these rules hold, there is a utility u such that lotteries are ranked by expected utility:

$$U(p) = \sum_{i=1}^n p_i u(z_i), \quad p \succeq q \iff U(p) \geq U(q).$$

Conversely, this way of ranking lotteries satisfies the four rules. Any other utility giving the same lottery rankings has the form

$$v(z) = au(z) + b, \quad a > 0,$$

called a *positive affine transformation*. It preserves the rankings because $\mathbb{E}[v] = a\mathbb{E}[u] + b$.

The theorem does not require concave or log utility, and it does not tell us what is fair across different users.

For fixed payoffs, x and $\log x$ both rank 9 above 5 above 1. But in part (d), linear utility makes A and B equally good ($5 = 5$), while log utility prefers A ($\log 5 > \log 3$). Thus a strictly increasing change of utility can keep the order of fixed payoffs but change the order of lotteries.

(f)

The reward hypothesis. An agent's goal can be described by a numerical reward, and the agent tries to maximize the expected total reward:

$$J(\pi) = \mathbb{E}_\pi \left[\sum_{t=0}^{T-1} R_{t+1} \right].$$

This is similar to VNM: both choose the option with the highest expected score. But VNM alone does not give a stationary Markov reward. Such a reward must use a fixed rule $r(s, a, s')$ based on the current transition. An overall score might also need past information that the state does not contain.

Compare the two schedules. J_{instant} adds the log utilities at each step. J_{average} first averages each user's service, then takes logs. For A, the two allocations are $(6, 3), (6, 3)$, so

$$J_{\text{instant}}(\pi_A) = 2(\log 6 + \log 3) = \log 324.$$

For B, the allocations are $(10, 1), (2, 5)$, so

$$J_{\text{instant}}(\pi_B) = \log 10 + \log 1 + \log 2 + \log 5 = \log 100.$$

A's average allocation is $(6, 3)$. B has the same average:

$$\left(\frac{10+2}{2}, \frac{1+5}{2} \right) = (6, 3).$$

Hence

$$J_{\text{average}}(\pi_A) = J_{\text{average}}(\pi_B) = \log 6 + \log 3 = \log 18.$$

The objectives are **not equivalent**: the first prefers A, while the second gives a tie.

A reward for average service. Give zero reward until the final step. At that step, give the utility of the average service:

$$R_{t+1} = 0 \quad (0 \leq t < T-1), \quad R_T = \log \bar{X}_1 + \log \bar{X}_2,$$

where $\bar{X}_i = \frac{1}{T} \sum_{t=0}^{T-1} X_i(t)$. Then

$$\mathbb{E}_\pi \left[\sum_{t=0}^{T-1} R_{t+1} \right] = \mathbb{E}_\pi [\log \bar{X}_1 + \log \bar{X}_2] = J_{\text{average}}(\pi).$$

To calculate this reward, keep track of each user's total service:

$$C_i(0) = 0, \quad C_i(t+1) = C_i(t) + X_i(t).$$

Add these totals and the time to the state:

$$\tilde{S}_t = (S_t, C_1(t), C_2(t), t).$$

The totals give $\bar{X}_i = C_i(T)/T$, and the clock tells us when to pay the final reward. If S_t already describes the environment's Markov dynamics, this larger state lets us use a fixed Markov reward rule without looking back through the whole history.

(g)

VNM says that, under its four rules, choices can be ranked by expected utility. RL uses a similar idea for a sequence of decisions: rank policies by expected total reward.

For example, in part (f) we want good average service over the whole episode. We set the final reward equal to $\log \bar{X}_1 + \log \bar{X}_2$ and all earlier rewards to zero. Maximizing expected total reward then gives exactly the wanted objective. We must also store the running service totals: VNM does not automatically tell us what the state or each step's reward should be.